

# The Factor $4/3$

Sphere Volume, Casimir Operators, Electromagnetic Mass, the Triple (3,4,5) and  $\xi$

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21 September 2026

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# The Factor $\frac{4}{3}$

## Abstract

The numerical factor  $4/3$  in the scaling parameter  $\xi = \frac{4}{3} \times 10^{-4}$  has so far been motivated at three places in the corpus: sphere volume (Docs. 006, 205), the Casimir operator of  $SU(3)$  and the Casimir effect (Doc. 324). This document adds two occurrences not previously recorded and orders all five by their common cause. First: the factor  $4/3$  in the electromagnetic mass of the electron, unexplained since Abraham (1902) and Lorentz (1904), arises because each of the three field components carries one third of the field energy — a consequence of three-dimensionality, not a computational error. Second: the Casimir values of the two smallest non-abelian groups,  $C_2(SU(2)) = 3/4$  and  $C_2(SU(3)) = 4/3$ , are reciprocals of each other;  $N = 2$  is the only case in which consecutive fundamental representations achieve this. Both values belong to the electron:  $3/4$  is its spin Casimir,  $4/3$  its mass factor. The Pythagorean triple  $(3, 4, 5)$ . The two factors  $3/4$  and  $4/3$  are reciprocals, and their product is 1. Where does this pair arise naturally? In the simplest right-angled triangle with integer sides: the triple  $(3, 4, 5)$ . It is not one among many; it is distinguished in four independent ways.

The triple  $(3, 4, 5)$  carries this reciprocal pair as  $\tan \alpha = 4/3$ ,  $\cot \alpha = 3/4$  and is, by Berggren's theorem, the root of all primitive triples. The angle  $\alpha = \arctan(4/3)$  does not occur among the shortest vectors of the lattices  $D_3$ ,  $D_4$ , but it does occur on the shell  $|v|^2 = 10$ : there lie  $(3, 1, 0, 0)$  and  $(1, 3, 0, 0)$  with  $\cos \alpha = 3/5$ . The cause is  $D_2 = (1 + i) \mathbb{Z}[i]$  and  $(1 + i)(2 \pm i) = 1 + 3i$ ,  $3 + i$ : the triple sits, via the factor  $(1 + i)$ , in the carrier of FFGFT. A further 3D occurrence, the enthalpy factor  $(u + p)/u = 4/3$  for isotropic pressure  $p = u/3$ , is added.

## Notation

### Epistemic markers:

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| <b>[B]</b>       | algebraically proved (Python verification script, exact arithmetic) |
| <b>[K]</b>       | numerically confirmed                                               |
| <b>[S]</b>       | structurally plausible, identification open                         |
| <b>[SETTING]</b> | motivated input, not a derivation                                   |
| <b>[E]</b>       | established external mathematics or physics                         |
| <b>[X]</b>       | checked and negative                                                |

| Symbol                | Meaning                                                                              |
|-----------------------|--------------------------------------------------------------------------------------|
| $\xi = 4/30\,000$     | dimensionless scaling parameter of FFGFT                                             |
| $C_2(G)$              | quadratic Casimir operator of the fundamental representation of $G$                  |
| $U, \mathbf{p}$       | field energy and field momentum of a moving charge                                   |
| $m_0 = U/c^2$         | electromagnetic mass from the rest energy                                            |
| $(3, 4, 5)$           | smallest primitive Pythagorean triple                                                |
| $\alpha$              | angle with $\tan \alpha = 4/3$ ; $\alpha \approx 53.13^\circ$                        |
| $D_3, D_4$            | root lattices in $\mathbb{R}^3$ (FCC) and $\mathbb{R}^4$                             |
| $D_2$                 | two-dimensional sublattice; coincides with the ideal $(1 + i) \mathbb{Z}[i]$         |
| $\mathbb{Z}[i]$       | ring of Gaussian integers $\{a + bi \mid a, b \in \mathbb{Z}\}$                      |
| $(2 + i)$             | Gaussian integer with $ 2 + i ^2 = 5$ ; its square is $(2 + i)^2 = 3 + 4i$           |
| $O(2, 1; \mathbb{Z})$ | integer Lorentz group of the form $x^2 + y^2 - z^2$ ; contains the Berggren matrices |
| Berggren matrices     | three integer $3 \times 3$ matrices generating the tree of all primitive triples     |
| $\epsilon_0$          | permittivity of vacuum                                                               |
| $e, c$                | elementary charge, speed of light                                                    |

## 1 Starting Point in the Corpus

The parameter  $\xi = \frac{4}{3} \times 10^{-4}$  carries two factors. The factor  $10^{-4}$  is treated in Doc. 006 as loop suppression and in Doc. 309 as an exponent setting; it is not the subject of this document. The factor  $4/3$  is motivated three times in the corpus:

1. **Sphere volume** (Docs. 006, 205):  $V = \frac{4}{3}\pi r^3$ ; the factor characterises the geometry of three-dimensional space.
2. **Casimir operator** (Doc. 324):  $C_2(\text{SU}(3)_{\text{fund}}) = \frac{9-1}{6} = \frac{4}{3}$ .
3. **Casimir effect** (Docs. 324, 009): the effective factor  $4/3$  in the Casimir energy between plates.

These three occurrences stand side by side in the corpus; Doc. 324 calls the agreement of (2) and (3) “no coincidence” without naming the common cause. This document names it and adds two occurrences.

## 2 Electromagnetic Mass: the Factor 4/3 since 1902

### The historical finding [E]

If the electron is modelled as a uniformly charged spherical shell of radius  $a$ , the field energy at rest is  $U = e^2/(8\pi\epsilon_0 a)$  and the associated mass is  $m_0 = U/c^2$ . If instead one computes the field momentum of the slowly moving shell, one obtains

$$\mathbf{p} = \frac{4}{3} \frac{U}{c^2} \mathbf{v}, \quad (1)$$

i.e. a momentum mass of  $\frac{4}{3}m_0$ . This factor first appeared in Abraham (1902) and Lorentz (1904). Poincaré (1906) introduced non-electromagnetic binding stresses that remove it computationally; Fermi (1922), Wilson (1936) and Rohrlich (1960) showed that a covariant definition of the four-momentum makes it vanish as well. The literature still disagrees on whether this explains the *cause* or merely redefines the *symptom* (Singal 2024; Universe 11(3):97, 2025).

**Why clarification around 1900 was so difficult.** The confusion had three mutually reinforcing sources.

*First: there was no special relativity.* Abraham and Lorentz worked in a framework where energy and momentum were still separate, independent quantities. Without  $E = mc^2$  and the four-momentum, a factor 4/3 between momentum and energy looks like a genuine contradiction — not like an artefact of a wrong separation.

*Second: the electron was new and its model undecided.* Thomson had discovered the electron in 1897. Whether it was a rigid sphere, a deformable shell, or a point mass was an open question. Within Lorentz's sphere model, the factor 4/3 was real and troublesome.

*Third: Poincaré's solution was the wrong solution.* The "Poincaré pressure" cancelled the factor computationally, but it was an ad-hoc assumption without physical foundation — and it obscured the actual cause.

After Einstein (1905) and Minkowski (1908) the problem could have been resolved cleanly: the factor 4/3 arises only when  $E$  and  $\mathbf{p}$  are integrated separately from the field, without accounting for the boundary terms of Maxwell's stress tensor. Rohrlich (1960) made this clear. What was never the subject of this debate: that  $4/3 = 2(1 - \frac{1}{3})$  is a geometric statement about the three-dimensionality of space. The question asked was "How do we get rid of the factor?" — rather than "Where does it really come from?"

### Where the factor comes from [E]

The calculation is short and exposes the cause. With  $\mathbf{B} = \mathbf{v} \times \mathbf{E}/c^2$  to first order, the momentum density is  $\mathbf{g} = \epsilon_0 \mathbf{E} \times \mathbf{B}$ , and

$$\mathbf{p} = \frac{\epsilon_0}{c^2} \int [\mathbf{v} E^2 - \mathbf{E}(\mathbf{E} \cdot \mathbf{v})] dV = \frac{\epsilon_0}{c^2} \mathbf{v} \int (E^2 - E_{\parallel}^2) dV. \quad (2)$$

For a spherically symmetric field,  $\langle \cos^2 \theta \rangle = 1/3$ , hence  $\int E_{\parallel}^2 dV = \frac{1}{3} \int E^2 dV$ , and with  $U = \frac{\epsilon_0}{2} \int E^2 dV$  it follows that

$$\mathbf{p} = \frac{\epsilon_0}{c^2} \cdot \frac{2}{3} \int E^2 dV \cdot \mathbf{v} = \frac{2}{3} \cdot \frac{2U}{c^2} \mathbf{v} = \frac{4}{3} \frac{U}{c^2} \mathbf{v}. \quad (3)$$

**The factor  $4/3$  is  $2 \cdot (1 - \frac{1}{3})$ .** It arises because in three-dimensional space each field component carries one third of the energy, and the component parallel to the motion does not contribute to the momentum. This is the same one-third split that sits in the sphere volume  $V = \frac{1}{3}rS$  (Archimedes). The factor is not an artefact of a wrong definition but a statement about the dimension of space.

### Connection to FFGFT [S]

In Doc. 006 the electron obeys  $E_e = 1/\xi_e$  with  $\xi_e = \xi \cdot f_e(1, 0, \frac{1}{2}) = \xi$ . In FFGFT the electron mass is therefore the reciprocal of  $\frac{4}{3} \times 10^{-4}$ . Classical electrodynamics has carried, since 1902, a factor  $4/3$  in exactly this quantity — the electron mass — whose cause is three-dimensionality. FFGFT does not attempt to define it away but places it as a geometric constant at the root. This is a structural agreement [S]; a derivation of one from the other is not available.

## 3 Casimir Values: $3/4$ and $4/3$ as a Reciprocal Pair

### The calculation [E]

The quadratic Casimir operator measures the total charge of a representation: it is the group-theoretic analogue of  $j(j+1)$  for angular momentum. For  $SU(2)$  and spin  $\frac{1}{2}$  it is  $\frac{1}{2}(\frac{1}{2} + 1) = \frac{3}{4}$  — the quantum spin number of the electron. For  $SU(3)$  in the fundamental representation (the three colour charges of quarks) it is  $\frac{4}{3}$  — the same factor as in the electromagnetic mass.

The quadratic Casimir operator of the fundamental representation of  $SU(N)$  is  $C_2 = \frac{N^2-1}{2N}$ :

| Group   | $C_2$ | physical location                           |
|---------|-------|---------------------------------------------|
| $SU(2)$ | $3/4$ | spin $\frac{1}{2}$ : $s(s+1) = \frac{3}{4}$ |
| $SU(3)$ | $4/3$ | colour charge; Doc. 324                     |

We have  $C_2(SU(2)) \cdot C_2(SU(3)) = 1$ . The verification script shows [B]: among all  $N \geq 2$ ,  $N = 2$  is the only case in which  $C_2(N) \cdot C_2(N+1) = 1$ . The two smallest non-abelian groups form a distinguished reciprocal pair.

### Both values belong to the electron [S]

The value  $3/4$  is the spin Casimir of the electron. The value  $4/3$  is, by §2, its electromagnetic mass factor and, in FFGFT, the numerator of its mass reciprocal  $1/\xi$ . Spin and mass of the electron thus carry the reciprocal pair  $(3/4, 4/3)$ . The fundamental formula  $\tilde{T} \cdot m = 1$  is a reciprocal structure; that the two electron invariants also form a reciprocal pair is structurally compatible, but not derived.

## 4 The Triple (3,4,5)

The reciprocal pair  $(3/4, 4/3)$  is the leg ratio of the smallest Pythagorean triple:  $\tan \alpha = 4/3$ ,  $\cot \alpha = 3/4$ ,  $\alpha = \arctan \frac{4}{3} \approx 53.13^\circ$ ,  $\sin \alpha = \frac{4}{5}$ ,  $\cos \alpha = \frac{3}{5}$ . The triple is not one among many; four properties distinguish it [E], all recomputed in the verification script [B]:

1. **Berggren root (1934).** Every primitive Pythagorean triple arises from  $(3, 4, 5)$  by a unique sequence of multiplications with three integer  $3 \times 3$  matrices. The three children of  $(3, 4, 5)$  are  $(5, 12, 13)$ ,  $(21, 20, 29)$ ,  $(15, 8, 17)$ ;  $(3, 4, 5)$  itself has no positive parent triple. It is the root of the tree.
2. **Arithmetic progression.**  $(3, 4, 5)$  is the only right triangle whose sides form an arithmetic progression:  $a^2 + (a + d)^2 = (a + 2d)^2 \Leftrightarrow (a - 3d)(a + d) = 0$ .
3. **Inradius 1.**  $r = (3 + 4 - 5)/2 = 1$ . The triples  $(5, 12, 13)$  and  $(8, 15, 17)$  have  $r = 2$  and  $r = 3$ ;  $(3, 4, 5)$  is the unit step.
4. **Gaussian integers.**  $(2 + i)^2 = 3 + 4i$ . The triple is the square of the smallest non-trivial Gaussian integer;  $\alpha$  is the angle of this squaring.

## The Five in the Triple and Its Connection to the Icosahedral Group [S]

The hypotenuse of the triple is 5, and 5 appears in the trigonometric values:  $\sin \alpha = 4/5$ ,  $\cos \alpha = 3/5$ . This 5 is not accidentally the same as the 5 in  $A_5$ .

The algebraic chain is as follows. The triple arises as the square of the Gaussian integer  $2 + i$ :  $(2 + i)^2 = 3 + 4i$  [B]. The Gaussian norm is  $|2 + i|^2 = 4 + 1 = 5$ . This number 5 is prime in  $\mathbb{Z}[i]$ : it splits as  $5 = (2 + i)(2 - i)$ , two conjugate Gaussian prime factors. The same prime 5 appears in the icosahedral group. The group  $A_5$  has order 60, and the fivefold rotation  $R_5$  has order 5. Its eigenvalue  $\zeta_5$  (fifth root of unity) lives in the finite field  $\text{GF}(81) = \text{GF}(3^4)$ , because  $5 \mid 80 = |\text{GF}(81)^*|$  [B]. In the same field  $\text{GF}(81)$ , via the subfield  $\text{GF}(9) \subset \text{GF}(81)$ , also lives the golden ratio  $\phi = (1 + \sqrt{5})/2$  (Docs. 364, 370).

The connection between the triple  $(3, 4, 5)$  and the icosahedral group  $A_5$  thus runs through the prime 5 and the field  $\text{GF}(81)$ :

$$(2 + i)^2 = 3 + 4i \implies |2 + i|^2 = 5 \implies 5 \mid |\text{GF}(81)^*| = 80 \implies \zeta_5 \in \text{GF}(81).$$

This connects the triple algebraically with the icosahedral part of the corpus (Docs. 285, 293, 367, 370). What this connection is not: a geometric identification. The angle  $\alpha = \arctan(4/3) = 53.13^\circ$  is not a rotation angle of  $A_5$  (which has  $72^\circ$ ,  $144^\circ$ ,  $120^\circ$ ,  $180^\circ$ ). The connection is algebraic, not geometric [S].

## 5 Common Cause and Limit

### What belongs together [S]

| Occurrence                           | Value    | Cause                 | Source         |
|--------------------------------------|----------|-----------------------|----------------|
| Sphere volume $V = \frac{1}{3}rS$    | 4/3      | one-third split in 3D | Docs. 006, 205 |
| EM mass, field momentum              | 4/3      | one-third split in 3D | new, §2        |
| Photon gas, $(u + p)/u$ at $p = u/3$ | 4/3      | one-third split in 3D | new, §5        |
| Casimir effect                       | 4/3      | 3 spatial directions  | Doc. 324       |
| $C_2(\text{SU}(3))$                  | 4/3      | $N = 3$               | Doc. 324       |
| $C_2(\text{SU}(2))$                  | 3/4      | $N = 2$               | new, §3        |
| $(3, 4, 5)$                          | 4/3, 3/4 | Berggren root         | new, §4        |

The enthalpy factor is the calculation of §2 in another language: an isotropic photon gas has pressure  $p = u/3$ , and the momentum of the moving system is  $\mathbf{p} = \frac{4}{3}(E/c^2)\mathbf{v}$  (von Laue) [E]. The first four rows share a common cause: the number 3 of spatial dimensions and the resulting one-third split. The Casimir rows have the number 3 as the rank of the

group. That  $SU(3)$  and three-dimensional space yield the same factor is, in FFGFT, not a coincidence but follows from the  $\mathbb{Z}_3$  triality on  $D_4$  (Doc. 365, layer 4) — this is the point at which the group number and the space number are already identified in the corpus.

### The angle $\alpha$ in the D lattice [B]

The  $D_n$  lattice is what crystallography calls the face-centred cubic (FCC) lattice: in  $\mathbb{R}^n$  all integer vectors whose coordinate sum is even. In FFGFT,  $D_4$  is the carrier lattice of the torus  $T^4$ ; its shortest vectors ( $|v|^2 = 2$ ) define its lattice constant. The question is: on which shells does the angle  $\arctan \frac{4}{3} = 53.13^\circ$  appear?

Among the shortest vectors ( $|v|^2 = 2$ ) of the  $D_3$  lattice (FCC) and the  $D_4$  lattice,  $\alpha = \arctan \frac{4}{3} = 53.13^\circ$  does not occur; both yield only  $60^\circ, 90^\circ, 120^\circ$  [X]. On the shell  $|v|^2 = 10$  it does occur:  $(3, 1, 0, 0)$  and  $(1, 3, 0, 0)$  lie in  $D_4$  (even coordinate sum), and  $\cos \angle = 6/10 = 3/5$  [B].

The cause is algebraic. The two-dimensional lattice  $D_2$  (even coordinate sum) is exactly the ideal  $(1 + i)\mathbb{Z}[i]$  of the Gaussian integers [B], and

$$(1 + i)(2 + i) = 1 + 3i, \quad (1 + i)(2 - i) = 3 + i, \quad (4)$$

with  $|1 + 3i|^2 = 10$ . The angle between  $2 + i$  and  $2 - i$  is  $\arccos \frac{3}{5} = \alpha$ ; multiplication by  $(1 + i)$  rotates both by  $45^\circ$  and leaves the angle unchanged. The triple  $(3, 4, 5)$ , which by §4 is the square of  $2 + i$ , therefore sits, via the factor  $(1 + i)$  — the generator of the D lattice — in the carrier geometry of FFGFT, on the first shell whose norm is not composed purely of  $|1 + i|^2 = 2$  and  $|2|^2 = 4$ .

The Berggren matrices are isometries of the quadratic form  $x^2 + y^2 - z^2$ , i.e. elements of the integer Lorentz group  $O(2, 1; \mathbb{Z})$  (Cha and Conceição 2023) [E]; the tree of all triples is thus a discrete hyperbolic structure on the same lattice.

### The Right Angle in FFGFT — a Further Thread [S]

That the  $90^\circ$  angle plays the key role in §2 has not yet been named explicitly in the text. The derivation shows it: the field component *parallel* to the direction of motion ( $0^\circ$ ) does not contribute to the field momentum; the two components *perpendicular* to it ( $90^\circ$ ) do. In three dimensions with three orthogonal directions, one falls out and two remain. That is  $2/3$ . Multiplied by 2 (from the cross-product structure of  $\mathbf{E} \times \mathbf{B}$ ) gives  $4/3$ . The geometric statement is:

$$\frac{4}{3} = 2 \cdot \left(1 - \frac{1}{3}\right) = 2 \cdot \frac{\text{number of directions perpendicular to motion}}{3}. \quad (5)$$

The  $90^\circ$  angle is not incidental but the cause. It also appears explicitly in the calculation:  $\langle \cos^2 \theta \rangle = 1/3$  follows from spherical symmetry, because this average over all angles is precisely what eliminates the right angle as a mean.

**The right angle in the  $D_4$  lattice.** Among the shortest vectors of  $D_4$  (even coordinate sum,  $|v|^2 = 2$ ) exactly the angles  $60^\circ, 90^\circ, 120^\circ, 180^\circ$  appear [B]. The  $90^\circ$  is thus a distinguished lattice angle, but it is not alone: it shares the space of lattice angles with  $60^\circ$  and  $120^\circ$ . Whether the  $90^\circ$  lattice angle plays a distinguished role in FFGFT beyond the others — for instance as the separation angle between the four torus directions, as an orthogonality condition between space and time, or as a condition for the projection  $T^4 \rightarrow \mathbb{R}^3 \times S^1$  — has not been investigated. That is the genuinely open question.

**The Fourier modes are orthogonal.** The three lepton modes  $v_0, v_1, v_2$  form an orthonormal basis:  $\langle v_j | v_k \rangle = \delta_{jk}$  **[B]**. Each pair of modes stands at  $90^\circ$  in Hilbert space. This orthogonality is the basic condition for the matrix element  $\langle v_0 | R_5 | v_1 \rangle$  to measure the mixing between two *perpendicular* states — and for the modulus squared  $|\langle v_0 | R_5 | v_1 \rangle|^2 = 2/9$  to be a transition probability in the sense of Born's rule. Without  $90^\circ$  in Hilbert space, no clean probability interpretation.

**2/3 and 4/3 as a bridge to the Koide quotient.** The Koide quotient is  $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$  **[E]**. The field-momentum formula from §2 gives  $\mathbf{p} = \frac{4}{3}(U/c^2)\mathbf{v}$ . Numerically, exactly:

$$\frac{4}{3} = 2 \cdot \frac{2}{3} = 2Q.$$

The factor  $4/3$  of the electromagnetic mass is twice the Koide quotient. This is a numerical identity — whether a deeper connection lies behind it, between the three-dimensionality factor of field physics and the mass-structure quotient of the leptons, is open **[S]**. It would be the most direct link of the Abraham-Lorentz problem to the lepton masses that the corpus currently contains: via the number  $2/3$ , which appears on the one hand as the Koide quotient, and on the other as the geometric fraction of transverse directions in three-dimensional space.

**The  $T^4$  torus and the four orthogonal circles.** The torus  $T^4$  in FFGFT carries four circle directions that in the standard setup stand pairwise at  $90^\circ$  — the simplest metric on the product torus  $S^1 \times S^1 \times S^1 \times S^1$ . These four  $90^\circ$  pairs generate the  $D_4$  lattice of winding numbers. The transition to spacetime  $T^4 \rightarrow \mathbb{R}^3 \times S^1$  picks *one* direction (the mass dimension that becomes time) and leaves the other three as spatial directions, pairwise at  $90^\circ$ . The three-dimensionality of space — that exactly *three* directions can stand perpendicular to a motion — is therefore not a free choice but follows from this projection step. The  $4/3$  factor is in this picture the signature of the symmetry that was left behind: the original four-dimensionality of the torus leaves in three-dimensional space exactly the three orthogonal directions that produce  $4/3 = 2 \cdot (1 - 1/3)$ .

Whether this reading —  $4/3$  as a trace of the projection  $T^4 \rightarrow \mathbb{R}^3 \times S^1$  — can be formally carried out in the corpus is the sharpest open question of this subsection **[S]**.

## What is not shown **[S]**

That the angle occurs in the carrier is shown. That it plays a *role* there — as a mode angle, a winding angle, or in the  $\mathbb{Z}_3$  identification — is not shown. The bridge currently reads: the same number  $4/3$  at six places with a common cause for four of them, and the associated angle on shell 10 of the carrier lattice. Whether shell 10 is distinguished in FFGFT is open.

## 6 Result

1. The factor  $4/3$  of the electromagnetic electron mass (Abraham 1902) is a three-dimensionality effect:  $4/3 = 2(1 - \frac{1}{3})$  **[E][B]**. It sits in the same quantity that appears in FFGFT as  $1/\xi$  **[S]**.
2.  $C_2(\text{SU}(2)) = 3/4$  and  $C_2(\text{SU}(3)) = 4/3$  are reciprocals;  $N = 2$  is the only such case **[B]**. Both values are invariants of the electron (spin, mass) **[S]**.
3.  $(3, 4, 5)$  is the Berggren root, the only triple in arithmetic progression, has inradius 1, and is the square of  $2 + i$  **[B]**. It carries the reciprocal pair as its legs.

4. The angle  $\arctan \frac{4}{3}$  is absent among the shortest vectors of  $D_3, D_4$  **[X]**, but occurs on the shell  $|v|^2 = 10$  because  $D_2 = (1+i)\mathbb{Z}[i]$  and  $(1+i)(2 \pm i) = 1 + 3i, 3 + i$  **[B]**. A role of this shell in FFGFT is open **[S]**.
5. The enthalpy factor  $(u+p)/u = 4/3$  at  $p = u/3$  is a further occurrence of the one-third split **[E]**.

**Verification script:** 2/python/Dok371\_Skripte/ffgft\_371\_faktor\_vier\_drittel.py — 28 checks, 28 passed (P1–P5, P7 positive; P6: negative control of the shortest vectors and positive finding on shell 10).

**Register (Doc. 190):** no entry — no earlier document is corrected; the occurrences in Docs. 006, 205, 324 remain valid and are supplemented.

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